Week 4 – Technical Presentation and Technical Writing (Lean Notes)

Lectures covered: L19 (how to make a technical presentation), L20 (technical writing). ⭐ = tested in the Week 4 assignment.

1. Concept cards (in the order taught)

Part A: Technical presentation (L19)

1. The lecture is itself an example

- A "talk about talks": 50 minutes, 5 topics (about 10 minutes each), about 35 slides. Everything below was demonstrated in the talk itself. Tick = good practice, cross = avoid.

2. Use an outline slide

- Start with an **outline** and **flash it again** between topics. It tells the audience what is coming, lets them focus on the part they care about, and shows where you are.
- End with **one summary slide** (two at most) so listeners can recall how the points connect.

3. Journal paper vs technical presentation

	Journal paper	Technical presentation
Nature	**one-time** for that data (you sign that it is not published elsewhere)	an **event**, **repeated** (group, conference, funding agency, school)
Structure	**rigid**: journal's word, page and figure limits and style	**flexible**: you reshape it for each audience
Review	**peer reviewed** (2–3 independent experts: new and valuable?)	**not** peer reviewed
Interaction	**not interactive** (you are absent when it is read)	**interactive** (questions during and after)
Audience	**no control** (school student to senior scientist), so keep it complete	**specific audience** each time, pitch to them
Life	**permanent record** that joins the body of knowledge (papers 100 years old are still read)	gone after the event

4. Two constraints: audience and duration ⭐

- **Audience background:** the same conference talk cannot go to a Class 10 school audience. Add time to explain terms and use concepts they know.
- **Why they are watching:** funding agency → what is unique and new; conference → new results or a new aspect nobody explored; school → general, exciting concepts.
- **Duration:** respect the time (it shows you value the audience's time and your own). Ask the host how much time you have. **Rule of thumb: about a slide a minute**; with confidence, fewer slides and more time per slide is better. Never flash slides fast.
- **Be agile:** know your work so well that a 40-minute slot cut to 25 minutes still carries the essence. Rough guide: **5 min ≈ 2–3 slides, 20 min ≈ 12 slides, 50–60 min ≈ 30–35 slides**. Do not keep the same 50 slides and press Enter fast. Content is the most important thing.

- Tiny number: 50-minute talk, 35 slides ≈ 1.4 minutes per slide, which is comfortable.
- Analogy: flashing the pages of a book does not teach the book. The audience needs **time with each piece of data**.

5. Connecting with the audience: do not overfill slides

- A busy slide (setup sketch + photos + micrographs + mechanical and electrical properties on one slide) is bad. **Empty space helps focus**. Split it into 4–5 slides, one concept per slide, in a logical flow.

6. Special effects and fonts

- Use special effects **judiciously**; too many **distract from the technical content**. A little (flashing one key item) can help.
- Stick to **one or two fonts** throughout.

7. Do not read from the slide

- Putting every word on the slide makes the **font tiny** (unreadable from 10 m) and the talk **monotonous**. Reading also means turning your back on the audience.
- Slide text = **pointers only** (for example 3 short-term goals, 3 long-term goals); then elaborate **conversationally**. **Face the audience and watch their eyes**: if they nod, speed up; if lost, elaborate.

8. Illustrations: choose the right type ⭐

- Types: **table, graph, photograph, drawing, schematic**. The illustration must **help the audience understand your point** (even if they disagree, they can say exactly what they disagree with).
- **Table** when you only need to show many exact values across samples. **Graph** when you want a trend such as a **maximum**; a string of numbers does not show a peak (16 x-y pairs in a table hide it, the same 16 points plotted show it at once).

9. Details of a table ⭐

- Always **headings and units** (for example 2θ in degrees, time in minutes).
- **Significant digits must match the measurement:** a ruler with 1 mm least count cannot give 19.967134 cm. Do not show absurdly many digits.
- Be **consistent**: the same number of significant digits down a column; if not possible, explain below the table why.

10. Details of a graph ⭐

- Label **both axes with the quantity and units**, and show the **numbers (values) on the axes**. A graph with only a shape and no values wastes the audience's time. The lecturer says it is better not to show the data at all than to show it without values.
- Add scale details that matter (example: current in nA versus voltage in mV **against SHE**, the reference electrode; the **scan rate**; a caption or title describing the setup). Use a **legend** if there are several curves.

11. Details of a photograph ⭐

- A photograph shows the **real object**, but remove **extraneous things** (a beaker, an instrument display in the background): put the object against a plain wall, floor or sheet of paper.
- Give an **overall photo plus a close-up** (fuel cell bicycle, then fuel cell stack).

12. Drawings and schematics

- A drawing can show the **atomic level** (graphene sheet with unit vectors a_1, a_2, and the chiral vector $Ch = n \cdot a_1 + m \cdot a_2$) or the **system level** (fuel cell with fuel and air in, DC power out). Label axes and vectors.

13. Level of complexity: photo vs drawing ⭐

- The Hubble telescope **photo** (arm, shuttle, earth, wiring, ports) distracts if you only want to say "it has solar panels". A **drawing** simplifies everything else and **highlights the panels**. Choose the simplest illustration that carries the one point.

14. Always show scale ⭐

- Without scale, the Taj Mahal looks 10 m tall; with people beside it (1.5–2 m each) it is about 75 m. Put in a **ruler, a familiar object or a scale bar** (carbon nanotube image: about 20 nm across, and the bar lets you measure the spacing between layers).

15. Summary slide

- Recap: outline, event vs paper, audience and duration constraints, agility, busy slides, effects, reading, choice of illustration, detail, complexity, scale. Close with "Thank you".

Part B: Technical writing (L20)

16. What a journal article is and how it gets published

- You find something **new** (new compared with all published work), write what you did, the results, their significance and context, and send it to a **journal** in your area. The journal sends it to **independent expert reviewers**. They recommend: publish as is, or publish after clarifications. You respond, then it is published (online first these days).
- Published journal articles become the **formal body of information** the world can use. Unpublished lab work is not formally available.
- You prove novelty by **comparing with earlier journal articles**. The number and quality of publications is **one measure** of scientific contribution (not the only one); top journals have higher standards.
- Reference book mentioned: **The Craft of Scientific Writing** by Michael Alley (Springer).

17. Technical writing is unlike other writing ⭐

Type	Style and purpose
Novels, mystery or suspense	keep you hooked to the end; **details are hidden**, the surprise comes last
Newspaper	day-to-day events, **urgency**, key facts, concise expert opinion
Magazine	more context, written for that magazine's audience
Journal article	**experts formally review** it to confirm it is new work; the aim is to **inform efficiently**

- Familiarity with other styles **misleads** students into writing a journal paper like a story. Result: a poor paper that gets returned.

18. Why do technical writing, and why it is hard ⭐

- **Why:** to get **formal credit** ("first to do this"); to **convey new results**; the results can influence **public policy** (for example, funding a cleaner fuel). It also helps the community keep track of work.
- **Why hard:** students do not know what a journal paper needs; the examples they know are written for a **different audience and purpose**; and they have **very high expectations of the first draft**. The first draft of your first 2–3 papers is usually poor technically. Frustration comes from not knowing what is being corrected. Reading papers **critically** improves your own writing.

19. Purpose of technical writing = inform and persuade, efficiently ⭐

- **Inform** = present your data (results). **Persuade** = discuss the data (argue what it means). Both must be done **efficiently**. Efficiency is the key idea behind every rule that follows.

20. Beginning: title, abstract, introduction ⭐

- **Title:** states the **area** and **what is unique**. “Rechargeable batteries” is too broad; “Mechanical failure of a specific rechargeable battery” names both.
- **Abstract:** sits right under the title and **gives away all the major findings**. This is the opposite of a novel. Students fear nobody will read on; but the aim is **efficient information**, not forcing people to read everything. The reader has **not read the paper**, so avoid undefined terms and abbreviations introduced later.
- **Introduction:** what your work is, **why it is important**, and what **background is needed** (for example a model the reader must know). Here you cite earlier papers (researcher x did this, y did that, z did this, and **this gap was neglected**), which shows your work is new. A **reference list** at the end holds all cited articles.

21. Middle: experimental details, results, discussion ⭐

- **Experimental details** (or the model, theory and implementation for modelling work): every setup and parameter, clear enough that **someone else can reproduce it**. Hide nothing, unless a patent is involved, and then say so.
- **Result** = a **factual statement of data** (“hardness rises as grain size falls”; “there is a maximum”). It has **no judgment** and cannot be argued with. Any technician can state it.
- **Discussion** = what the result **means**: your argument or reasoning (for example two competing parameters, one raising and one lowering the value, so a maximum appears). It is based on science but different people can differ, so you have more freedom.
- **Organising the middle: chronological** (experiment 1, 2, 3), **spatial** (layer by layer, outer shell of a reactor inward, temperature rising), or **following a variable** (for example viscosity of a fluid along its path). Not suspense.

22. Ending: conclusions ⭐

- Concisely restate the **main results**. Unlike the abstract, the reader **has read the paper**, so more flexible language is fine. You may add a **future perspective** (for example study operating conditions of the better fuel). The abstract does not usually carry future work.

23. Language ⭐

- **Simple, straight, clear sentences**. No flowery language. **Avoid very long sentences**: by the end the reader should not have forgotten the start or have to reread. **Avoid phrases like “it is obvious”** (improper and arrogant). Short sentences are also less ambiguous.

24. Illustration choice in a paper

- Choose the illustration that conveys the point **efficiently**; this is where you can be creative. A **summary table** of all experiments lets the reader see the range at a glance and refer to it while reading.
- Same rule as in the talk: a schematic of a satellite with two solar panels beats a cluttered photograph; a graph shows a maximum better than a 4 × 4 table of numbers.
- The lecturer ends by asking you to **read papers critically** and expect your first drafts to improve.

2. Top traps (most likely first)

1. **Title:** it must give the **area and the unique point**. A vague title such as “Battery design and its impact” fails on both counts.
2. **Abstract gives all results away**; it is not a teaser. Abstract (reader has not read the paper, so no later abbreviations) differs from conclusions (reader has read it, future perspective allowed).
3. **Result vs discussion:** result = fact, no judgment, cannot be argued; discussion = interpretation, opinion based on science.
4. **Middle of document strategies:** chronological, spatial, following a variable. **Suspense is a fiction technique** and is not used.

5. **Tables vs graphs:** table for **exact values**, graph for **trends and maxima**.
6. **Table details:** headings, units, significant digits that match the instrument, consistent digits.
7. **Graph details:** axis labels with units and values; a reference (for example vs SHE) and the scan rate when relevant; a legend for many curves.
8. **Photographs and images:** only the needed detail, no distracting background, and **always a scale** (scale bar, ruler or familiar object). Photo vs drawing depends on complexity.
9. **Do not read from slides**; do not overfill; one or two fonts; effects only when they help.
10. **Time:** about a slide a minute, adjust to the slot, never flash slides.
11. **Journal paper vs presentation:** the paper is one-time, rigid, peer reviewed, not interactive, with an uncontrolled audience; the presentation is a repeatable, flexible, interactive event with a known audience.
12. **Do not copy novel-style writing**; avoid long sentences and "it is obvious"; first drafts are poor, so expect revisions.

3. Assignment questions with solutions

Q1. The title of a paper is "Battery design and its impact". This title (select all that apply) *(Q1)* - A. Identifies the area of research clearly B. Indicates what is unique about the research C. Does not identify the area of the research clearly D. Does not indicate what is unique about the research

Solution (cards 20, 5): "Battery" and "impact" are too broad (which battery, impact on what?), and nothing says what is new. So A and B are false. **Answer: C and D.**

Q2. In images: *(Q2)* - A. High magnification implies high resolution B. High magnification does not necessarily imply high resolution C. Low magnification implies high resolution D. High resolution implies low magnification

Solution (outside the lectures, see table below): Magnification is how large the picture is drawn; resolution is how finely detail can be told apart. Enlarging a blurry image adds size, not detail. **Answer: B.**

Q3. Tables are preferred over graphs: *(Q3)* - A. Always B. Never C. Only when exact values need to be indicated D. Only when trends in data need to be indicated

Solution (cards 8, 9): Table for exact values; trends and maxima are better shown by a graph. **Answer: C.**

Q4. The length of a document (select all that apply) *(Q4)* - A. Indicates how much you know B. Indicates how little you know C. Does not indicate anything about your knowledge D. Does not indicate anything about the reader's knowledge

Solution (outside the lectures): The lectures judge writing by **efficiency**, not length; length measures neither the author's nor the reader's knowledge. **Answer: C and D.**

Q5. The title of a paper should indicate the area of work as well as what is unique about the work. *(Q5)* - A. True B. False

Solution (card 20): Stated exactly in the lecture. **Answer: A (True).**

Q6. In scientific documents sentences should be relatively short and to the point. *(Q6)* - A. True B. False

Solution (card 23): Long sentences make the reader reread. **Answer: A (True).**

Q7. In an article, the image of a car engine must (select all that apply) *(Q7)* - A. Show the level of detail required for the discussion in the article B. Have a marker bar indicating scale C. Be in colour and not in black and white D. Be in black and white and not in colour

Solution (cards 13, 14, 24): Show just the detail needed (complexity), and always give scale. Colour is not a rule either way. **Answer: A and B.**

Q8. Strategies one can use in the middle of a document include (select all that apply) *(Q8)* - A. Chronological strategy B. Spatial strategy C. Following a variable D. Suspense strategy

Solution (card 21): The lecture lists the first three; suspense belongs to novels. **Answer: A, B and C.**

Q9. While working with electricity only the voltage poses a danger to the users. *(Q9)* - A. True B. False

Solution (outside the lectures; links to Week 3 safety): The harm is done by the **current** through the body, which depends on voltage and resistance, so voltage alone is not the whole story. **Answer: B (False).**

Q10. The sentence "Pouring rain results in clogged drains, and slippery roads. These clogged drains spill further water on to the roads." Use of the word "these" is ambiguous. *(Q10)* - A. True B. False

Solution (outside the lectures; follows the "no ambiguity" idea of card 23): "These" is followed by the noun "clogged drains", so it clearly points back to the drains. It would be ambiguous only if used alone (for example "These spill water") with two possible earlier items. **Answer: B (False).**

Not taught directly in the Week 4 lectures (answered from general knowledge):

Question	Fact used
Q2	magnification (size) is independent of resolution (detail)
Q4	document length measures neither the writer's nor the reader's knowledge
Q9	current through the body is the danger
Q10	a pronoun followed by its noun is not ambiguous